Causal Validation and Behavioral Grounding of Moral Representation Directions in Language Models

Orion Reblitz-Richardson*

Abstract

Linear probes can find directions that separate moral from neutral text, but a probe cannot distinguish genuine moral representation from surface regularity. Probing shows that information exists in a model's representations; it cannot show the model computes with it.

We extract directions for the six foundations of Moral Foundations Theory in OLMo-2 1B and test whether the model computes with them during generation. Three methods converge. Ablating a direction specifically impairs generation of that foundation's content: the model needs it. Injecting a direction at low amplitude produces foundation-specific behavioral shifts with a dose-response curve characteristic of real feature manipulation: the model responds to it. Projection-based classification predicts which foundation a novel text activates, well above chance on held-out data and exceeding 80% on targeted prompts.

Sparse autoencoder features independently recover the probe subspace at over three times the chance rate. One finding complicates the picture: real-world moral vignettes overwhelmingly activate sanctity/degradation regardless of intended foundation, reflecting how the model encodes moral transgression rather than a deficiency of the directions.

These directions are not just readable but writable into models with predictable effect.

1 Introduction

Companion paper (Reblitz-Richardson, 2026) established that language models develop structured moral representations: six foundation-specific probe directions exhibit integration geometry (effective dimensionality 5, mean pairwise cosine ≈ 0.22–0.27), and these directions are stable across extraction methods, dataset sizes, and text registers. But probe directions are correlational. A direction that separates moral from neutral text in activation space may reflect a confound (sentence length, formality, topic) rather than a genuine moral feature. Three questions remain open:

1. **Causal relevance.** Does the model *use* these directions during generation? If we ablate a foundation's direction, does the model lose access to that specific moral concept? If we inject a direction, does the model shift toward that foundation's content?
2. **Behavioral grounding.** Do the directions predict which moral foundation a novel text activates? Representational geometry tells us how the model *organizes* moral knowledge; behavioral benchmarking tests whether that organization is *functionally accessible* for moral reasoning.
3. **Mechanistic correspondence.** Do the supervised probe directions correspond to features the model discovers on its own? Sparse autoencoders (SAEs) learn unsupervised features from general text. If morally selective SAE features align with probe directions, the directions are not artifacts of the probing procedure but reflect native model structure.

*Distiller Labs. Correspondence to Distiller Labs <orion@orionr.com>.

Preprint.

We address all three questions using the same OLMo-2 1B model and 240-pair moral probing dataset from Reblitz-Richardson (2026). Section~4.1 presents direction ablation and steering injection experiments. Section~4.3 reports projection-based foundation classification on held-out, external, and causal evaluation stimuli. Section~4.4 compares SAE feature geometry with probe directions. Section~5 synthesizes the three forms of evidence and discusses implications for representation engineering of moral concepts.

2 Related Work

2.1 Causal methods in mechanistic interpretability

Activation patching (Meng et al., 2022, Vig et al., 2020) and causal tracing (Geiger et al., 2021) establish whether identified features are functionally relevant by measuring behavioral changes under targeted interventions. Direction ablation (projecting out a direction from hidden states) tests necessity: does the model need this direction? Steering injection (adding a scaled direction to hidden states) tests sufficiency: can this direction control model behavior? Arditi et al. (2024) demonstrated that refusal behavior is mediated by a single direction, establishing the approach we apply to moral foundations. Turner et al. (2023) introduced activation addition for steering, showing that concept directions can modulate generation at inference time.

2.2 Representation engineering

Zou et al. (2023) proposed extracting concept directions via paired-difference PCA and using them for model control. Marks and Tegmark (2024) extended this to multi-dimensional concept spaces. Our work differs in validating directions extracted by three independent methods (probe weights, mean-difference, LEACE) and giving causal evidence that the directions are load-bearing for the specific moral concepts they encode.

2.3 Sparse autoencoders for feature discovery

SAEs decompose neural network activations into sparse, interpretable features (Bricken et al., 2023, Cunningham et al., 2023). If supervised probe directions align with unsupervised SAE features, this is evidence that the directions correspond to natural features of the model's computation rather than artifacts of the probing procedure. Prior work has used SAEs primarily for safety-relevant features (Templeton et al., 2024); we apply them to structured moral concepts.

2.4 Moral reasoning in language models

Reblitz-Richardson (2026) established the geometric structure of moral representations that this paper validates causally. The Moral Foundations Vignettes (Clifford et al., 2015) offer an independent external stimulus set for behavioral benchmarking.

3 Methods

3.1 Model and dataset

All experiments use OLMo-2-0425-1B (16 layers, 2048 hidden dim) with mean-difference directions extracted from the 240-pair moral probing dataset (40 pairs per MFT foundation, 192 train / 48 test). Directions are normalized to unit length. We use mean-difference directions rather than probe-weight directions because mean-difference directions capture more of the shared moral-salience component (mean pairwise cosine 0.41 vs. 0.22 for probe weights at layer 0; Reblitz-Richardson 2026, §4.13), making them better suited for causal interventions that target the full moral representation rather than the maximally discriminative direction. The choice matters: SAE subspace overlap is 15.5% for mean-difference vs. 8.2% for probe-weight directions (Section 4.4.3). See Reblitz-Richardson (2026) for dataset construction and direction extraction.

3.2 Causal validation methods

3.2.1 Direction ablation

For each foundation f and layer ℓ, we register a forward hook that projects out the foundation direction $\mathbf{d}_f^{(\ell)}$ from the hidden state:

$$\mathbf{h} \leftarrow \mathbf{h} - (\mathbf{h} \cdot \mathbf{d}_f^{(\ell)})\, \mathbf{d}_f^{(\ell)}$$

We then measure the change in log-probability of target continuations from a 48-prompt causal evaluation set (8 prompts per foundation, with labeled target and off-target continuations).

Metrics. For each (ablated foundation, layer):

- *On-target* Δ: mean log-prob change on prompts whose target foundation matches the ablated direction.
- *Off-target* Δ: mean log-prob change on prompts from other foundations.
- *Specificity*: on-target Δ minus off-target Δ. Negative specificity indicates that ablation specifically harms the target foundation.

3.2.2 Steering injection

For each foundation f, layer ℓ, and amplitude $\alpha \in \{1, 2, 5, 10, 20\}$, we add a scaled direction to the hidden state:

$$\mathbf{h} \leftarrow \mathbf{h} + \alpha\, \mathbf{d}_f^{(\ell)}$$

and measure log-probability changes on the same evaluation set. The dose–response curve across α values distinguishes genuine feature manipulation (specificity at low α, saturation at high α) from noise injection (monotonic degradation).

3.2.3 Causal evaluation prompts

The 48-prompt evaluation set comprises three formats: 24 completion prompts (4 per foundation) with a sentence stem and 3–4 candidate continuations, 12 forced-choice prompts (2 per foundation), and 12 natural prompts (2 per foundation). Each prompt has labeled target and off-target continuations with foundation labels, enabling measurement of foundation-specific log-probability changes under ablation and injection.

Prompt construction. Prompts were hand-authored to activate specific MFT foundations. Completion prompts use sentence stems where the target continuation is a single word or short phrase whose foundation loading is unambiguous (e.g., a care/harm stem ending in `she felt compelled to'' with target`help’’ and off-target `ignore'' and`report’’). Forced-choice prompts present two continuations from different foundations. Natural prompts use open-ended stems that implicitly prime a specific foundation. All prompts were reviewed to ensure that (a) the target foundation is the most natural continuation, (b) off-target continuations span at least two other foundations, and (c) no prompt relies on surface lexical cues (the foundation name or its synonyms do not appear in the stem). The full prompt set with all continuations and foundation labels is included in the code repository.

3.3 Behavioral grounding methods

3.3.1 Projection-based classification

For each input text, we collect the last-token residual stream activation at target layers (4, 8, 12), project onto all six foundation directions, and average across layers. The foundation with the highest projection is the predicted label. We also test a *debiased* variant that subtracts the mean projection across all six foundations before classification, removing the shared moral-salience component.

3.3.2 Evaluation sets

1. **Held-out test set** (48 pairs, 8 per foundation): internal validation from the probing dataset.
2. **Moral Foundations Vignettes** (30 items, 5 per foundation): curated from Clifford et al. (2015), offering external validation with established MFT stimuli independent of our dataset pipeline.

3. **Causal evaluation prompts** (48 prompts): cross-validation with the causal experiments, testing whether directions that are causally relevant also predict foundation identity.

3.4 Sparse autoencoder analysis methods

3.4.1 SAE training

We train a ReLU sparse autoencoder (16,384 latent dimensions, $8\times$ expansion) on 2M token activations from the C4 corpus (Raffel et al., 2020) at layer 8, using L1 sparsity regularization ($\lambda = 0.005$). The decoder columns are constrained to unit norm. The pre-encoder bias is initialized to the mean activation.

3.4.2 Moral feature identification

We encode 192 moral and 192 neutral training sentences through the trained SAE and compute per-feature selectivity: the mean activation difference between moral and neutral inputs. Top-k features (ranked by |selectivity|) are compared with probe directions via:

- **Individual alignment**: cosine similarity between each feature's decoder column and each foundation direction.
- **Subspace overlap**: fraction of probe direction variance captured by the top-k SAE feature subspace (via SVD projection).

3.4.3 Random baseline

To establish significance of subspace overlap, we compute a null distribution: 1,000 iterations of projecting probe directions onto 100 random unit vectors in $\mathbb{R}^{2048}$ and recording mean membership.

4 Results

4.1 Direction ablation is foundation-specific

Ablating a foundation's direction from the residual stream specifically reduces log-probability of that foundation's continuations while leaving other foundations largely unaffected. Specificity (on-target Δ minus off-target Δ) is negative for all foundations at layers 8 and 12, confirming that each direction carries information specific to its foundation.

Layer dependence. Specificity increases with depth. At layer 4, mean specificity across foundations is -0.16; at layer 8 it is -0.39; at layer 12 it is -0.63. The strongest effects are sanctity at layer 12 (specificity $= -1.64$, on-target $\Delta = -1.68$ nats) and liberty at layer 12 (-0.64). This pattern (ablation most damaging in later layers) is consistent with the model relying more heavily on moral representations as they approach the output.

Foundation heterogeneity. Sanctity is the most causally load-bearing direction at all layers (specificity -0.42 at layer 4, -0.48 at layer 8, -1.64 at layer 12), while care and fairness show the weakest effects at early layers (near-zero specificity at layer 4). At layer 12, all six foundations show negative specificity, confirming that the directions are not redundant: each carries unique information that the model uses for generation.

4.2 Steering injection shows dose–response specificity

At low amplitude ($\alpha = 1$), injecting a foundation direction into the residual stream at layer 8 produces a positive on-target boost for four of six foundations (mean $+0.15$ nats across foundations) with near-zero off-target effect (mean $+0.03$), yielding positive specificity for four foundations and near-zero for two (care and fairness).

Dose–response curve. As α increases:

- $\alpha = 1$–2: foundation-specific boost. On-target log-probability increases for most foundations, off-target is near baseline. Sanctity and loyalty show the strongest specific effects.

- $\alpha = 5$: moderate amplification. Mean specificity rises to $+0.88$ at layer 8. All foundations except fairness show specificity > 0.2.
- $\alpha = 10$–20: non-specific amplification. Both on-target and off-target log-probabilities increase, but on-target increases *more*, so specificity grows further (mean $+2.34$ at $\alpha = 10$, layer 8). The large positive specificities at high α reflect genuine directional information rather than noise.

This dose–response pattern distinguishes the moral directions from random directions: noise injection would produce monotonic degradation at all amplitudes, not the low-α specific boost and high-α amplification we observe.

Layer dependence. Layer 4 shows the strongest injection effects at all α values (mean specificity $+0.95$ at $\alpha = 5$), while layer 12 shows the weakest ($+0.33$). This pattern is complementary to ablation: later layers are more sensitive to direction *removal* (ablation), while earlier layers are more responsive to direction *addition* (injection), suggesting that moral information flows from early-layer encoding to late-layer utilization.

4.3 Behavioral grounding: projection predicts foundation identity

Projection-based 6-way classification (debiased) achieves:

Dataset	Accuracy	Chance
Causal prompts	83.3% (40/48)	16.7%
Held-out test set	70.8% (34/48)	16.7%
Moral Foundations Vignettes	33.3% (10/30)	16.7%

4.3.1 Test set performance

Per-foundation accuracy is uneven: liberty and sanctity (87.5%) perform best, followed by care (75.0%), fairness and authority (62.5%), and loyalty (50.0%). The confusion matrix shows that loyalty errors scatter to sanctity, consistent with a shared binding-foundations component; fairness errors are more distributed across foundations.

4.3.2 External validation: Moral Foundations Vignettes

The MFV accuracy of 33.3% (above the 16.7% chance baseline) masks a systematic pattern: the confusion matrix shows that sanctity/degradation dominates classification regardless of the true foundation. All five sanctity items are correctly classified, but 20 of 25 non-sanctity items are also classified as sanctity.

This is a property of the stimuli interacting with the direction geometry. The MFV items are descriptions of witnessing morally transgressive acts (`You see a boy kicking a puppy''), and many involve purity/disgust reactions that activate the sanctity representation. A loyalty violation (You see a team member sharing` confidential information' ') also involves betrayal of sacred trust; an authority violation ("You see a student cursing at a teacher' ') also involves degradation of a sacred relationship.

This co-activation is consistent with the integration geometry: the shared moral-salience component (mean pairwise cosine ≈ 0.22–0.27 between foundation directions) means that stimuli activating *any* foundation will partially activate *all* foundations. For MFV stimuli specifically, the sanctity/purity dimension receives disproportionate activation because harm-witnessing scenarios carry implicit purity content. Debiasing (subtracting the mean projection) does not resolve this because the sanctity co-activation is genuine, not an artifact of the shared component.

4.3.3 Causal prompt performance

The 83.3% accuracy on causal evaluation prompts (which were designed for targeted foundation activation, not harm witnessing) confirms that the directions are behaviorally predictive when stimuli are foundation-specific. Per-foundation accuracy is uniformly high (62.5–100%), with loyalty achieving perfect classification (100%).

4.4 SAE features partially recover moral subspace

4.4.1 Training results

The layer-8 SAE achieves L0 = 1,932 (11.8% of 16,384 features active per token) and FVU = 0.285 (71.5% variance explained) after 3 epochs on 2M tokens. The moderate FVU reflects limited training scale; production SAEs typically train on 100M+ tokens.

4.4.2 Moral selectivity

Only 4 of 16,384 features have moral selectivity $|s| > 0.1$ (mean activation on moral minus neutral text). Moral information is distributed across many features rather than concentrated in a few, consistent with the high effective dimensionality of moral representations found in Reblitz-Richardson (2026).

4.4.3 Subspace overlap

The top-100 morally selective SAE features, despite individually showing no alignment with probe directions (0% of features with $|\cos| > 0.2$), collectively span a subspace that captures 15.5% of mean-difference direction variance. The random baseline is $100/2048 = 4.88\%$, yielding a ratio of $3.17\times$ random. Probe-weight directions show weaker overlap (8.2%, $1.67\times$ random), consistent with probe-weight directions capturing partially different aspects of the moral subspace than mean-difference directions.

This result has two interpretations. First, even a partially trained SAE discovers features that overlap with supervised moral directions at $3.2\times$ the chance level, suggesting these directions reflect genuine structure in the model's representations rather than probing artifacts. Second, the overlap is modest (15.5%), indicating that either (a) the SAE needs substantially more training to decompose moral features cleanly, or (b) moral representations are inherently distributed across many low-selectivity features that resist separation by current SAE methods.

5 Discussion

5.1 Three converging lines of evidence

The causal, behavioral, and mechanistic results converge on a single conclusion: the moral probe directions identified in Reblitz-Richardson (2026) are genuine features of the model's computation, not artifacts of the probing procedure.

Direction ablation establishes *necessity*: the model needs these directions to generate foundation-specific content (mean specificity -0.63 at layer 12). Steering injection establishes *sufficiency*: adding these directions shifts model behavior toward the target foundation, with a dose–response curve characteristic of real feature manipulation. Behavioral grounding establishes *functional accessibility*: the directions predict which foundation a novel text activates, at accuracy levels far above chance (83.3% on causal prompts, 70.8% on held-out test data). SAE analysis establishes *mechanistic correspondence*: unsupervised feature discovery partially recovers the same subspace that supervised probing identifies ($3.2\times$ random baseline).

No single line of evidence would be conclusive. Ablation could reflect a generic disruption effect; injection could be noise-driven; behavioral accuracy could reflect confounds; SAE overlap could be coincidental. But taken together, the convergence across independent methods with different assumptions and failure modes gives strong evidence for the reality of the moral direction structure.

5.2 The sanctity saturation phenomenon

The MFV results expose a property of real-world moral stimuli that laboratory-controlled probing datasets obscure: witnessing moral transgressions of *any* kind preferentially activates the sanctity/degradation representation. This is not a deficiency of the probing directions but a genuine feature of moral cognition as encoded by the model. Sanctity/purity concepts involve reactions to violations of sacred norms and bodily integrity (Graham et al., 2013), and many moral transgressions implicitly involve degradation of sacred relationships, trust, or dignity.

This finding connects to the sanctity anomaly identified in Reblitz-Richardson (2026): sanctity is the most robust foundation in dense models but the most fragile in MoE models (6.2× ratio). The MFV saturation pattern suggests that sanctity directions carry particularly broad moral-salience information, making them both highly activated by diverse stimuli and causally load-bearing (sanctity has the strongest ablation specificity at all layers).

This has practical implications for steering applications: injecting a non-sanctity foundation direction into a context that already contains transgression content will contend with strong sanctity co-activation. Effective foundation-specific steering may require *suppressing* the sanctity component while boosting the target foundation, requiring a two-direction intervention rather than a single injection.

5.3 Layer-dependent causal roles

Ablation and injection reveal complementary layer-dependent patterns:

- **Ablation specificity increases with depth** (strongest at layer 12), indicating that later layers rely more heavily on foundation-specific information for generation.
- **Injection at layer 4 produces stronger effects** than layer 12, suggesting that earlier layers are more receptive to direction addition, since injected information has more layers to propagate through before reaching the output.

These patterns suggest a processing hierarchy: earlier layers encode moral content in a form that is more malleable (responsive to injection) but less causally connected to output (weak ablation), while later layers encode it in a form that is more rigid (weak injection response) but directly drives generation (strong ablation).

5.4 Toward a steering fitness function

The results suggest a concrete fitness function for evaluating moral directions as steering targets:

1. **Ablation specificity** < -0.3 at the target layer (direction is causally load-bearing).
2. **Injection specificity** > 0 at $\alpha = 1$ (direction can produce specific behavioral shifts).
3. **Behavioral accuracy** $> 50\%$ on held-out data (direction is functionally discriminative).
4. **SAE subspace overlap** $> 1.5\times$ random baseline (direction corresponds to native features).

Sanctity, liberty, and loyalty directions pass all four criteria. Fairness and care pass criteria 1 and 3 but show weaker injection specificity at low α, suggesting these foundations have more distributed representations that are harder to selectively amplify. Authority passes all criteria at moderate levels.

5.5 Limitations

SAE training scale. The 2M-token SAE achieves only 71.5% variance explained (FVU = 0.285), and only 4 of 16,384 features show moral selectivity > 0.1. The $3.2\times$ random subspace overlap, while above chance, is modest; a skeptical reader could question whether this level of overlap is distinguishable from structured but non-moral correlations in the SAE features. We treat this as a lower bound: production-scale SAE training (100M+ tokens, targeting FVU < 0.05) would likely increase both feature quality and moral direction overlap. We estimate that overlap exceeding $5\times$ random ($\sim 25\%$) with > 10 selective features would constitute strong evidence of moral feature recovery; achieving this likely requires ≥ 50M training tokens based on scaling trends in the SAE literature.

Single model. All experiments use OLMo-2 1B. Generalization to larger models and different architectures is untested.

Causal evaluation set size. The 48-prompt evaluation set (8 per foundation) limits statistical power for per-foundation claims. The direction of all effects is consistent, but effect sizes should be interpreted with appropriate uncertainty.

Steering injection confounds. At high α, injection amplifies model behavior broadly. The positive specificity at high α reflects *relative* on-target amplification, not selective control. The therapeutically relevant regime is $\alpha = 1$–2.

6 Conclusion

We have presented three independent forms of evidence that the moral probe directions identified in Reblitz-Richardson (2026) are genuine features of OLMo-2 1B's computation. Direction ablation establishes necessity (specificity -0.63 at layer 12); steering injection establishes sufficiency (specific boost at $\alpha = 1$–2 with dose–response amplification); and sparse autoencoder analysis establishes mechanistic correspondence ($3.2\times$ random subspace overlap). Behavioral benchmarking confirms that these directions are functionally accessible, achieving 70.8% accuracy on held-out test data and 83.3% on causal evaluation prompts.

The sanctity saturation phenomenon (sanctity/degradation activation dominating classification of real-world moral stimuli regardless of target foundation) is not a limitation of the directions but a genuine property of moral representation that connects to the sanctity anomaly from Reblitz-Richardson (2026) and has implications for steering applications. Effective moral steering may require multi-direction interventions that suppress sanctity co-activation while boosting the target foundation.

These results transform the moral geometry from a descriptive finding into a tool for representation engineering. The directions are not just *readable* from the model; they are *writable* into the model with predictable behavioral effects. Future work should test generalization to larger models, refine the SAE analysis with production-scale training, and develop the multi-direction steering strategies that the sanctity saturation phenomenon motivates.

Acknowledgments and Disclosure of Funding

This work made extensive use of Anthropic's Claude (the Claude Code agent on Opus 4.6, 4.7, and 4.8) for code scaffolding, experimental scripts, and prose drafting. The author retains responsibility for experimental design, all scientific claims, and final wording.

References

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. *arXiv preprint arXiv:2406.11717*, 2024. URL https://arxiv.org/abs/2406.11717.

Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E. Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. URL https://transformer-circuits.pub/2023/monosemantic-features.

Scott Clifford, Vijeth Iyengar, Roberto Cabeza, and Walter Sinnott-Armstrong. Moral foundations vignettes: A standardized stimulus database of scenarios based on moral foundations theory. *Behavior Research Methods*, 47(4):1178–1198, 2015. doi: 10.3758/s13428-014-0551-2.

Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. *arXiv preprint arXiv:2309.08600*, 2023. URL https://arxiv.org/abs/2309.08600.

Atticus Geiger, Hanson Lu, Thomas Icard, and Christopher Potts. Causal abstractions of neural networks. *Advances in Neural Information Processing Systems*, 34, 2021.

Jesse Graham, Jonathan Haidt, Sena Koleva, Matt Motyl, Ravi Iyer, Sean P. Wojcik, and Peter H. Ditto. Moral foundations theory: The pragmatic validity of moral pluralism. In *Advances in Experimental Social Psychology*, volume 47, pages 55–130. Academic Press, 2013.

Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *arXiv preprint arXiv:2310.06824*, 2024. URL https://arxiv.org/abs/2310.06824.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. *Advances in Neural Information Processing Systems*, 35, 2022.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.

Orion Reblitz-Richardson. How language models organize competing moral frameworks. *arXiv preprint*, 2026. Companion paper.

Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, Brian Chen, Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jones, Hoagy Cunningham, Nicholas L. Turner, Callum McDougall, Monte MacDiarmid, C. Daniel Freeman, Theodore R. Sumers, Edward Rees, Joshua Batson, Adam Jermyn, Shan Carter, Chris Olah, and Tom Henighan. Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. *Transformer Circuits Thread*, 2024. URL https://transformer-circuits.pub/2024/scaling-monosemanticity.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023. URL https://arxiv.org/abs/2308.10248.

Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart Shieber. Investigating gender bias in language models using causal mediation analysis. *Advances in Neural Information Processing Systems*, 33, 2020.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023. URL https://arxiv.org/abs/2310.01405.

Appendices

Supplementary material.

A Full Ablation Tables

Tables 1 through 3 report the complete ablation results for all six foundations at layers 4, 8, and 12. Each cell shows the mean change in log-probability (nats) under direction ablation.

Ablated foundation	On-target Δ	Off-target Δ	Specificity
Care	-0.00	$+0.00$	-0.00
Fairness	$+0.01$	-0.00	$+0.01$
Liberty	-0.23	$+0.00$	-0.24
Loyalty	-0.22	-0.03	-0.19
Authority	-0.13	$+0.00$	-0.14
Sanctity	-0.45	-0.04	-0.42
Mean			$\mathbf{-0.16}$

Table 1: Direction ablation at layer 4.

Ablated foundation	On-target Δ	Off-target Δ	Specificity
Care	-0.12	-0.05	-0.07
Fairness	-0.37	-0.10	-0.27
Liberty	-0.56	-0.01	-0.55
Loyalty	-0.54	-0.02	-0.52
Authority	-0.49	-0.05	-0.45
Sanctity	-0.44	$+0.04$	-0.48
Mean			$\mathbf{-0.39}$

Table 2: Direction ablation at layer 8.

Ablated foundation	On-target Δ	Off-target Δ	Specificity
Care	-0.52	-0.08	-0.44
Fairness	-0.21	-0.06	-0.15
Liberty	-0.71	-0.07	-0.64
Loyalty	-0.60	-0.12	-0.48
Authority	-0.44	-0.05	-0.39
Sanctity	-1.68	-0.03	-1.64
Mean			$\mathbf{-0.63}$

Table 3: Direction ablation at layer 12. Sanctity shows by far the strongest ablation effect.

B Steering Dose-Response Tables

Tables 4 through 6 report specificity (on-target Δ minus off-target Δ) for all six foundations at each injection amplitude and layer.

Foundation	α=1	α=2	α=5	α=10	α=20
Care	-0.08	-0.29	$+0.12$	$+2.55$	$+3.50$
Fairness	-0.06	$+0.02$	$+0.22$	$+1.28$	$+1.43$
Liberty	$+0.17$	$+0.11$	$+1.10$	$+3.31$	$+2.29$
Loyalty	$+0.35$	$+0.83$	$+1.01$	$+2.85$	$+4.68$
Authority	$+0.19$	$+0.57$	$+1.32$	$+2.95$	$+4.30$
Sanctity	$+0.26$	$+0.68$	$+1.96$	$+5.04$	$+6.47$
Mean	$+0.14$	$+0.32$	$+0.95$	$+3.00$	$+3.78$

Table 4: Steering injection specificity at layer 4.

Foundation	α=1	α=2	α=5	α=10	α=20
Care	-0.00	$+0.00$	$+0.51$	$+2.40$	$+4.41$
Fairness	-0.05	$+0.09$	$+0.20$	$+0.31$	$+0.60$
Liberty	$+0.17$	$+0.24$	$+0.97$	$+2.32$	$+2.54$
Loyalty	$+0.27$	$+0.55$	$+1.03$	$+2.19$	$+3.73$
Authority	$+0.04$	$+0.21$	$+0.82$	$+2.50$	$+3.76$
Sanctity	$+0.27$	$+0.52$	$+1.74$	$+4.31$	$+5.57$
Mean	$+0.12$	$+0.27$	$+0.88$	$+2.34$	$+3.44$

Table 5: Steering injection specificity at layer 8.

C Behavioral Confusion Matrices

D SAE Feature Details

D.1 Training hyperparameters

D.2 Per-foundation subspace overlap

D.3 Random baseline distribution

The analytical expectation for projecting a random unit vector in $\mathbb{R}^{2048}$ onto a random 100-dimensional subspace is $100/2048 = 4.88\%$. The observed mean-difference overlap of 15.5% (3.17× baseline) confirms that the morally selective SAE features capture genuine structure shared with supervised probe directions.

E Reproducibility

E.1 Hardware and software

All experiments ran on a single MacBook Pro with Apple M4 Pro (24 GB unified memory) using the MPS backend. Software versions: Python 3.13, PyTorch 2.7, Transformers 4.49.

E.2 Runtime

E.3 Model checkpoint

Model	Repo	Revision	Used for
OLMo-2 1B	`allenai/OLMo-2-0425-1B`	`main`	All experiments

The model is a base (non-instruct) checkpoint loaded in float16 precision with `low_cpu_mem_usage=True`.

Foundation	α=1	α=2	α=5	α=10	α=20
Care	+0.16	+0.33	+0.75	+1.54	+3.36
Fairness	+0.03	+0.05	+0.04	+0.14	+0.86
Liberty	+0.02	−0.05	−0.11	−0.07	+0.44
Loyalty	+0.06	+0.14	+0.41	+0.79	+1.66
Authority	+0.04	+0.10	+0.27	+0.73	+1.89
Sanctity	+0.20	+0.37	+0.61	+1.17	+2.16
Mean	+0.08	+0.15	+0.33	+0.72	+1.73

Table 6: Steering injection specificity at layer 12. Layer 12 shows weaker injection effects than earlier layers, complementing its stronger ablation effects.

True ↓ / Pred →	Care	Fair	Lib	Loy	Auth	Sanc
Care	**6**	0	0	0	0	2
Fairness	1	**5**	0	0	1	1
Liberty	0	0	**7**	0	0	1
Loyalty	0	0	0	**4**	0	4
Authority	0	0	0	1	**5**	2
Sanctity	0	0	0	0	1	**7**

Table 7: Confusion matrix for debiased projection-based classification on the held-out test set (48 pairs, 8 per foundation). Overall accuracy: 70.8%.

E.4 Random seeds

Experiment	Seed(s)	Where set
Probing dataset split	42	`deepsteer/datasets/pipeline.py`
SAE training	torch default	SAE training script
Ablation/injection noise	deterministic (no stochastic component)	—

E.5 Causal evaluation prompt set

The 48-prompt evaluation set is hand-authored (Section 4.1.3) and version-controlled in the code repository at `papers/4_causal_validation/outputs/causal_eval_prompts.json`. The file contains all prompts, continuation texts, foundation labels, and target/off-target annotations.

E.6 SAE training hyperparameters

Parameter	Value
Latent dimensions	16,384 (8× expansion)
Activation function	ReLU
L1 sparsity coefficient (λ)	0.005
Training tokens	2M (C4 corpus)
Target layer	8
Epochs	3
Pre-encoder bias	initialized to mean activation
Decoder column constraint	unit norm

E.7 Reproducibility notes

Direction extraction. Mean-difference directions are computed as $\mathbf{d}_f = \bar{\mathbf{a}}_{\text{moral}} - \bar{\mathbf{a}}_{\text{neutral}}$ (normalized to unit length) on the 192-pair training split. No optimization is involved.

True ↓ / Pred →	Care	Fair	Lib	Loy	Auth	Sanc
Care	**2**	0	0	0	0	3
Fairness	0	**2**	0	0	0	3
Liberty	0	0	**1**	0	0	4
Loyalty	0	0	0	0	0	**5**
Authority	0	1	0	0	0	4
Sanctity	0	0	0	0	0	**5**

Table 8: Confusion matrix for Moral Foundations Vignettes (30 items). Sanctity dominates classification: 20 of 25 non-sanctity items are classified as sanctity.

True ↓ / Pred →	Care	Fair	Lib	Loy	Auth	Sanc
Care	**7**	0	0	1	0	0
Fairness	0	**5**	1	1	1	0
Liberty	1	0	**7**	0	0	0
Loyalty	0	0	0	**8**	0	0
Authority	0	0	0	0	**7**	1
Sanctity	0	0	0	0	2	**6**

Table 9: Confusion matrix for causal evaluation prompts (48 prompts). Foundation-targeted stimuli yield uniformly high accuracy (83.3%) with no sanctity dominance, confirming that the MFV pattern reflects stimulus properties rather than direction artifacts.

Ablation and injection. Both interventions use PyTorch forward hooks that modify the residual stream in-place during a single forward pass with `torch.no_grad()`. Ablation projects out the direction component; injection adds a scaled direction vector. Results are deterministic given fixed inputs and directions.

Code availability. All experiment scripts, the probing dataset, the causal evaluation prompt set, and figure generation code are available at https://github.com/deepsteer/deepsteer.

Parameter	Value
SAE width	16,384
Expansion factor	$8\times$
Hidden dim	2,048
Training tokens	2M (C4)
Batch size	4,096
Learning rate	3×10^{-4}
L1 coefficient	5×10^{-3}
Epochs	3
L0 (final)	1,932
FVU (final)	0.285

Table 10: SAE training configuration and final metrics.

Foundation	Mean-diff overlap	Probe-weight overlap
Care	14.5%	8.6%
Fairness	15.9%	9.2%
Liberty	16.3%	7.8%
Loyalty	15.6%	8.8%
Authority	18.9%	8.2%
Sanctity	11.8%	6.4%
Mean	**15.5%**	**8.2%**
Random baseline	4.9%	4.9%
Ratio	$\mathbf{3.17}\times$	$\mathbf{1.67}\times$

Table 11: Per-foundation subspace overlap between top-100 morally selective SAE features and probe directions. Random baseline is $100/2048 = 4.88\%$.

Experiment	Model	Wall time
Direction ablation (Section 4.1)	OLMo-2 1B	$\sim$5 min
Steering injection (Section 4.2)	OLMo-2 1B	$\sim$10 min
Behavioral benchmarking (Section 4.3)	OLMo-2 1B	$\sim$3 min
SAE training (Section 4.4)	OLMo-2 1B	$\sim$20 min
SAE moral feature analysis	OLMo-2 1B	$\sim$2 min